



Alex A. DeOliveira

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Profile

Senior Controls & Systems Engineer and Computer Scientist with 25+ years of multidisciplinary engineering experience, combining hands-on technical execution with initiative ownership spanning Industrial Automation, SCADA, Railway Systems (CBTC), Electronics, and systems integration. Knowledgeable in Artificial Intelligence and Machine Learning through postgraduate coursework, with limited applied experience in supporting AI-driven automation in R&D environments. Committed to integrity-centered collaboration, accountability, and operational ownership, delivering secure, high-availability, real-time industrial automation systems that advance modernization and long-term operational excellence. Fluent in English, Portuguese, and Spanish

Technical Expertise

- SCADA & Infrastructure: Real-time systems (RedHat Linux, Windows Server), network configuration (routers, switches, firewalls), cybersecurity awareness, 24/7 system reliability
- Industrial Automation: PLC programming, HMI, SCADA (iFIX), OPC, Profibus, TCP/IP, industrial Ethernet, controls integration, variable frequency drives, soft-starters, inverters, COBOL programming, algorithm & programming logic
- Hands-On Engineering: Control panels, VFDs, vision systems, soft starters, motors, pumps, conveyors, encoders, relays, sensors, instrumentation, electrical-mechanical troubleshooting & assembly
- Documentation & Compliance: AutoCAD schematics, UL/CE support, FAT/SAT, technical manuals, signal analysis
- AI & Applied Machine Learning (Postgraduate Studies – California Institute of Technology | University of Michigan): AI engineering, RAG workflows, vector embeddings, neural network fundamentals, LLM, applied ML for automation optimization, etc

Education

MICHIGAN STATE – USA/ONLINE, PRO EDU Applied Gen AI Specialization (in progress)

CALTECH – California/USA – **PosGrad** – Artificial Intelligence & Machine Learning (2024/2025)

FUNDAÇÃO GETÚLIO VARGAS – Brazil – MBA - Project Management (2017/2019)

SOUTHERNPOLYTECHNIC STATE UNIVERSITY, Marietta, GA – B.S. Electrical Engineering (2004/2009)

BROOKDALE COMMUNITY COLLEGE, Lincroft, NJ – **A.A.S. in Computer Sciences** (1999/2002)

Professional Experience

ATS/SP SCIENTIFIC/AB&C Engineering – Warminster PA Jan 2022 -Present

Controls Systems Engineer, R&D

•Engineered specialized Software & Controls projects, including PLC code development, iFIX/SCADA commissioning, and electrical troubleshooting for lyophilization systems. Delivered 24/7 remote/field support and authored comprehensive technical documentation, schematics, and operational manuals..

WABTEC BRAZIL – Brazil – Various Cities 2013-2019

• Systems & Application Engineer, Team Coordinator

Leadership role in executing Brazil's first pilot CBTC project implementation on Locomotives & Railway Auxiliary Vehicles. Field surveys, hands-on installation, quality assurance, software & hardware troubleshoot. Created 300 plus docs such as electrical schematics, surveys, installation manuals, signal analyses, checklists.

ULTRASONIC MOLECULAR WELDING – Shelton CT 2010-2013

• Controls Engineer/Assistant R&D

PLC/HMI programming and design of control panels for Ultrasonic Welding Automation. AutoCAD Schematics, PCB Designing, Research/Creation of Support Documentation for UL and CE certification.

ELECTRONIC MACHINES ASSOCIATES – Norcross, GA 2008-2010

• Field/Lab Engineer

Installation, repair and maintenance of Variable Frequency Drives. Troubleshooting of board level components, power electronics, control systems and electrical motor wiring.

OPTELIAN ACCESS NETWORKS – Marietta, GA 2006-2008

• Network Design Specialist

Overall technical support + CAD drawings for network designs of CWDM & DWDM applications.

McNEAL WIRELESS SERVICES – Kennesaw, GA 2006

• Associate RF Engineer

Conducted CW and PMV testing as well as signal data processing & propagation in antenna cell towers.

HEWLETT-PACKARD/CINGULAR - End of Lease Roll-Off Project, Cedartown, GA 2006

Network & Technical Support (A Team Effort Project w/ HP, Cingular and TCM technicians)

Performed hardware installation and network protocol mapping.

NOR-RAL AEROSPACE COMPOSITES INC. – Canton, GA 2006-2007

• IT/Database Administration

Assisted with the overall day-to-day network and database operations of a mid-size company.

REYNOLDS AND REYNOLDS – Duluth, GA 2005 – 2006

• Specialist ODS Programmer

Programmed & maintained Pick Basic programs for the finance and insurance of automobile dealerships.

BRAYSON'S HOME – Acworth & Lawrenceville, GA 2001 - 2005

Assistant Superintendent

Conducted daily inspections of various levels of home building.

STAR TECHNICAL INSTITUTE – Ocean Township, NJ 2000 – 2001

• Computer Instructor

Prepared students for A+ Hardware/Software Certification by utilizing theoretical and hands-on approaches

BROOKDALE COMMUNITY COLLEGE – Lincroft, NJ Spring 2000

• Math Tutor

Assisted faculty staff by providing individual instruction to college students in Advanced Algebra, Trigonometry, Calculus I, II, and III.

